Cross-Margin on Zodial

A Guide for Kamino and Marginfi Users

You've deposited into three different markets on Kamino before just to avoid a liquidation you knew was coming. Not because your portfolio was actually at risk, but because your collateral was locked in the wrong market to help.

That's the problem Zodial is built to remove.

One Portfolio, One Health Score

On Kamino and Marginfi, every position you open lives in its own isolated market. Your SOL collateral backing a SOL borrow can't help cover a separate TSLAx borrow sitting in a different market, even if that SOL collateral has plenty of room to spare. Each market asks its own narrow question: is this specific position healthy. Your actual financial position across the whole account is invisible to the protocol.

Zodial asks a different question: is your portfolio healthy. All your deposits back all your borrows at once. One health score, not five.

What Isolation Actually Costs You

Take a simple two-position portfolio: $1,000 in staked SOL and $800 in TSLAx, each backing a separate borrow in an isolated market.

If TSLAx drops to $600, the isolated model liquidates that position immediately, even though the SOL side of the portfolio hasn't moved and has plenty of unused capacity. The staked SOL collateral is sitting right there, healthy, and structurally forbidden from helping.

Run the same scenario in a unified model and the portfolio stays covered, because the SOL headroom gets pulled in automatically. Push it further, drop TSLAx all the way to $300, and the unified portfolio still holds. That's not a marginal improvement. That's the difference between getting liquidated on a dip and not.

Not All Pairs Get the Same Leash

Here's the part isolated protocols get wrong in the other direction too: they either ignore correlation entirely (Aave's flat collateral-weighted LTV) or only account for it in a handful of predefined pairings (Kamino's e-mode). Zodial prices every asset pair individually, based on how much that pair's relative price has actually moved against each other historically.

The mechanism: for any collateral/borrow pair, Zodial tracks the maximum historical price divergence over a rolling window. That number becomes the Liquidation Threshold (LT) for the pair. Loan-to-Value is set at 98% of LT, and the liquidation penalty is 50% of LT. Correlated, stable pairs get a longer leash. Volatile or historically divergent pairs stay tight, because the math says they should.

Here's what that looks like with real published thresholds:

Asset	Max LT	What It Means
USDC	97.0%	Near-maximal borrowing power, minimal historical divergence risk
SOL	91.1%	High capital efficiency, standard crypto volatility priced in
SPY	93.3%	Broad index exposure, tight spread reflects low divergence
XAUT	89.8%	Gold-backed, conservative but still efficient
NVDA	87.0%	Single-stock volatility pulls the threshold down
TSLA	86.5%	Higher historical divergence, tighter LT accordingly

> **Worth flagging:** these are the published single-asset max LTs, the ceiling for that asset against its best-case counterparty. The actual pairwise LT for a specific combination, say NVDAx against USDC specifically versus NVDAx against SOL, may sit lower depending on the correlation between that exact pair. That pairwise figure isn't public yet. Treat anything pair specific in this doc as illustrative until confirmed.

Why Can't Kamino Just Add This

Fair question if you're coming from Kamino. The honest answer is structural, not a matter of Kamino not getting around to it yet.

E-mode boosts LTVs for narrow, predefined asset groupings, it doesn't generalize to arbitrary portfolio compositions. Isolated markets are built around single-position health checks, there's no mechanism in that architecture to ask whether your entire collateral stack covers your entire debt stack. And even if you wanted to bolt that on, solving for the optimal capital flow across a large multi-asset portfolio is a computationally heavy optimization problem, the kind of thing that's genuinely infeasible to run on-chain given blockchain compute limits.

Zodial's answer to that last problem is the actual unlock: an off-chain solver computes the optimal capital flow and generates a certificate proving the portfolio is covered, then the smart contract just verifies that certificate rather than redoing the math itself. That's the piece that makes portfolio-wide cross-margin possible at all.

Walking Through a Mixed Position

Say you deposit $1,000 USDC and open a long NVDAx position through a loop, borrowing against your stablecoin collateral to gain leveraged exposure.

The protocol doesn't just check whether this one NVDAx position is covered. It runs the capital flow calculation across your whole account, and the Playground reflects this directly, showing something like *"$X*

capital used, covered with LT 0.87 by $Y." That LT 0.87 is NVDA's pairwise threshold against whatever you're borrowing, and the $Y figure is the exact collateral value backing that specific borrow, with anything left over shown as unused headroom, not locked away.

Now say NVDAx drops. Health degrades on that side of the portfolio, but before anything gets touched, the system checks whether headroom elsewhere, unused capacity on other deposits, can absorb the move. Only when the portfolio-wide health score actually crosses its threshold does liquidation logic activate at all. This is the same mechanic that saved the SOL/TSLAx example above, just running live on your position.

(Note: exact figures above are illustrative, built to show the mechanic rather than reflect a live quote.)

How Liquidation Actually Works

Trigger	What Happens	Why
Portfolio health score crosses 1.0 (not any single position's isolated threshold)	The off-chain solver computes the minimum liquidation needed to restore health, not a blanket close-out	The same maximum capital flow math that proves a portfolio is covered can also calculate the smallest corrective action, so liquidation is targeted, not total
A specific pair is the one dragging health down	That pair gets liquidated first, other healthy positions in the same portfolio are left alone	Liquidation follows the same pairwise logic as borrowing power, it isolates the actual problem rather than penalizing the whole account
Liquidation executes	A penalty equal to 50% of that pair's LT is applied	This is the same LT-derived formula used for borrowing power, kept consistent across the whole risk model

Partial liquidation, in other words, isn't a special exception carved out for user protection. It falls directly out of the same certificate based math that makes cross-margin borrowing work in the first place. One mechanism, two uses.

One More Layer: Closed Market Assets

Tokenized stocks like NVDAx and TSLAx track real equities that don't trade 24/7. Zodial handles this by maintaining the peg through direct price alignment during market hours, and using the last official closing price when the underlying market is closed. To prevent someone from exploiting a stale close-of-market price, the LTV thresholds for these assets are built from historical stock data with a safety buffer already priced in. You're protected from the exact kind of after-hours price attack that a naive oracle setup would be vulnerable to.

The Honest Caveat

Everything mechanical in this doc, the LT/LTV/penalty formulas, the isolated versus unified comparison, the off-chain proof system, the closed market handling, comes straight from what Zodial has published. The specific numbers used in the walkthroughs above are illustrative, built to demonstrate how the mechanic behaves, not live quotes. The one thing genuinely unconfirmed is the pairwise LT for individual collateral/borrow combinations beyond the single-asset max already public. That gets updated the moment the team confirms it.